

The Weasley Clock

Design notes for the family build: where the idea comes from, how it is put together, and what is left to do.

Hardware plan	Weasley Clock Hardware Plan (architecture, parts, costs, build phases)
Order checklist	Weasley Clock Order Checklist
Circuit diagram	Weasley Clock Circuit Diagram (pin-level wiring)
3D model (interactive)	Weasley Clock CAD viewer (case, dial, hands, screens and the motor stack)
Code and CAD files	https://github.com/Austin-Hunter1/HunterClock

All of these links are public and open without any login. The site home page is <https://austin-hunter1.github.io/HunterClock/>.

1. Inspiration

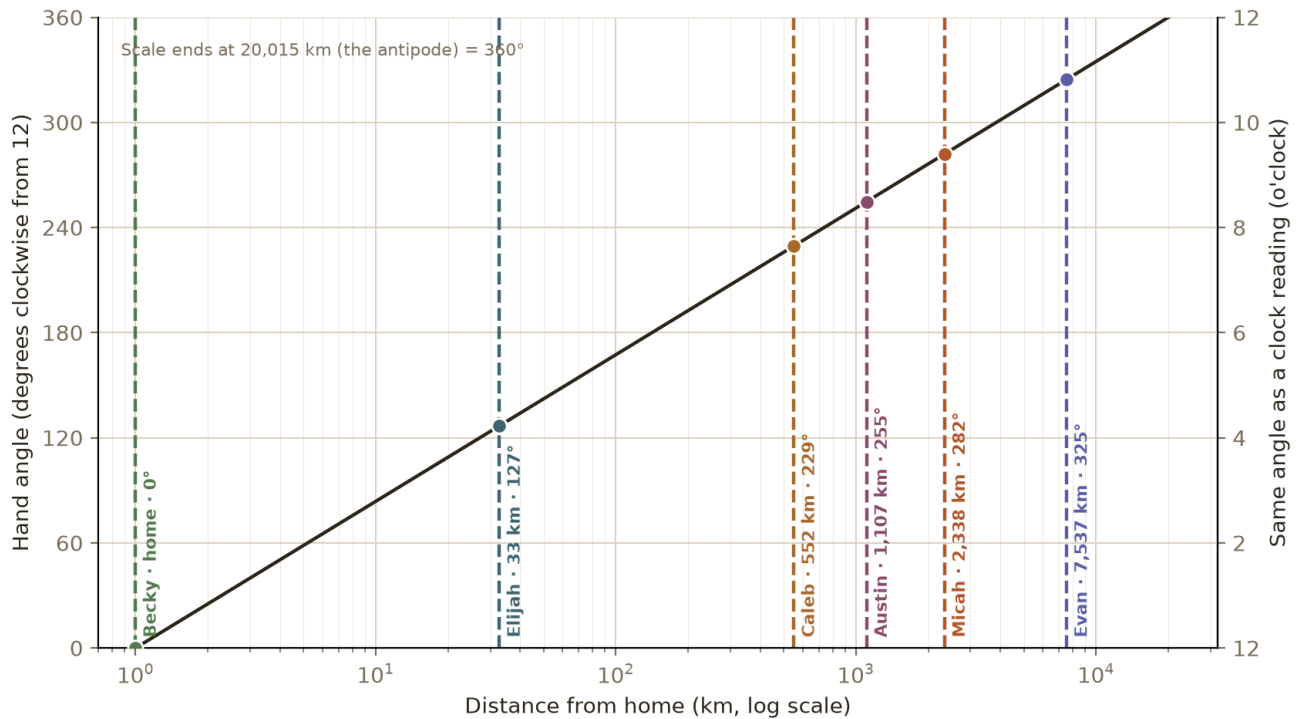
The Weasley Clock comes from the Harry Potter books. The Weasley family keeps a clock at the Burrow with no numbers on it. Instead, each family member has a hand of their own, and the dial is marked with places and situations such as home, school, work and travelling, down to "mortal peril" on the worst days. The hands move as the family moves, so anyone in the house can see where everyone is at a glance.

The wizarding world adds a second idea: pictures move. A portrait or a photograph is a small living loop of the person in it.

Our clock keeps both ideas and changes one. **The hands show how far each person is from home**, not which category they are in: 12 o'clock means home, and the farther away someone is, the farther their hand sweeps around the dial. **The moving pictures sit on the outside of the clock face**, in round screens beside the dial, each looping a short real video of its person. The case is not a family heirloom. It is a grandfather clock that Caleb found (a Howard Miller triple-weight moon-phase model), which we are gutting of its original movement, weights and pendulum.

2. How distance becomes a hand angle

Distance comes from the Haversine formula, clamped to a minimum of 1 km. A $\log^{10}$ scale then maps it so that 1 km sits at 0° (12 o'clock) and the antipode, about 20,015 km, sits at 360° (back to 12). The scale is logarithmic so a hand moves visibly for a 30 km difference and for a 7,000 km one. In log space the mapping is a straight line, and each person is a vertical line at their distance. The source is birthday_clock.m; the Python scripts and CAD model use the same data.



Hand angle against distance from home. The right axis reads the same angle as a clock time, at 30 degrees per hour.

Person	Location	Distance (km)	Hand angle
Becky	Minnetonka, MN	1 (home)	0.0°
Elijah	St. Paul, MN	33	126.8°
Caleb	Wheaton, IL	552	229.5°
Austin	Boulder, CO	1,107	254.8°
Micah	Corvallis, OR	2,338	281.9°
Evan	Warsaw, PL	7,537	324.5°

Angles are measured clockwise from 12 o'clock.

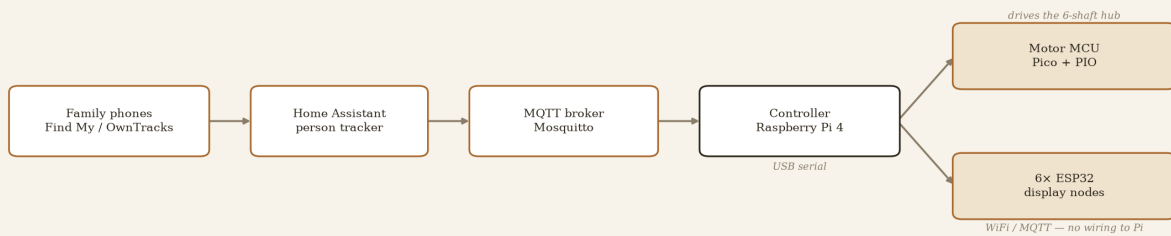
3. Mechanical design

One hub, six hands

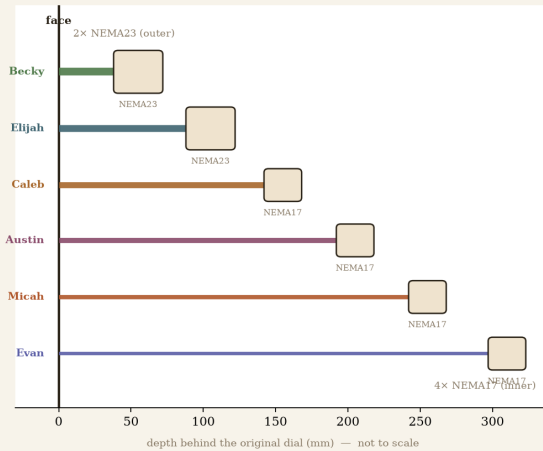
All six hands pivot from one centre point, so the hands sit on concentric shafts. Six hollow-shaft stepper motors are chained nose to tail along the depth axis behind the dial. Each motor's shaft passes through the hollow bores of the motors behind it, and a telescoping brass tube carries it forward to its hand. The result is the single-pivot look of the show clock, with the whole trunk cavity of the case available for the stack.

The Weasley Clock — System Overview

Location → distance → hand angle, plus the display pipeline that runs beside it

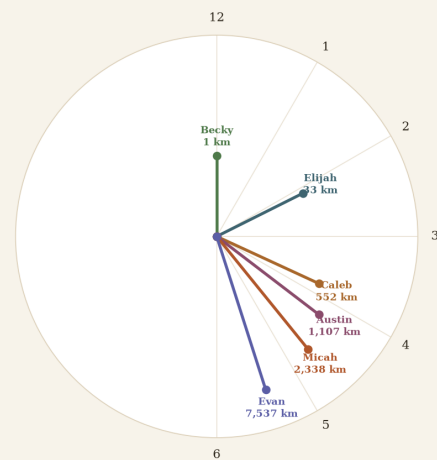


Hub cross-section (side view, schematic)



Distances via Haversine from Minnetonka, MN · logio scale, 1 km → 12:00, 20,015 km (antipode) → 6:00

Front face — all 6 hands, live angles



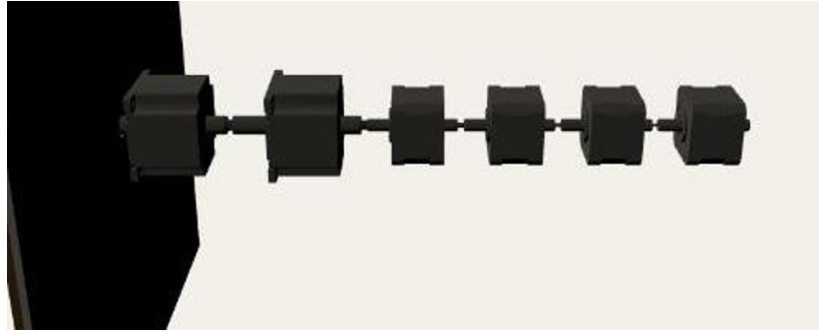
System overview and hub cross-section from the hardware plan (visualize_hardware_plan.py). The face in the third panel uses the older 180-degree mapping.

Motors and shafts

Level	Hand	Motor	Bore / shaft OD	Tube to the hand
1 (front)	Becky or Evan	NEMA23 hollow-shaft, 23HS18-2004H	8 mm / 12 mm	None; the hand mounts on the motor shaft
2	Evan or Becky	NEMA23 hollow-shaft, 23HS18-2004H	8 mm / 12 mm	9/32 in (7.1 mm) brass
3	Elijah	NEMA17 hollow-shaft, 17HS13-1504H	4 mm / 8 mm	1/4 in (6.4 mm) brass
4	Caleb, Austin or Micah	NEMA17 hollow-shaft, 17HS13-1504H	4 mm / 8 mm	1/8 in (3.2 mm) brass
5	(same three)	NEMA17 hollow-shaft, 17HS13-1504H	4 mm / 8 mm	3/32 in (2.4 mm) brass
6 (rear)	(same three)	NEMA17 hollow-shaft, 17HS13-1504H	4 mm / 8 mm	1/16 in (1.6 mm) brass

Becky, Evan and Elijah move the most, so they get the three stiffest shafts. Tubes are K&S telescoping brass. Between tubes 3 and 4 the diameter drops by more than one standard step, because tube 4 must clear the 4 mm bore of the level 3 motor. That one interface needs a small guide bushing; every other interface is a standard consecutive size. Which of Becky and Evan takes

level 1, and the order of Caleb, Austin and Micah, are not yet decided.



The motor stack seen from the side with the case hidden: two NEMA23 motors in front, four NEMA17 behind, shafts passing through each other's bores.

The hardware plan estimated a stack depth of 300 to 320 mm. In the CAD, using the vendor models, it comes to about 420 mm, from the front tip of the first motor shaft to the rear tip of the last, because each hollow shaft sticks out of both ends of its motor (25 mm front and 58 mm rear on the NEMA23, 16 mm and 43 mm on the NEMA17) and 4 mm is left between motors for the couplings. That is deeper than the plan assumed, so the case measurements matter. A jam anywhere in the stack can bind every hand behind it; we accepted that in exchange for the single-pivot look.

4. Screens and moving pictures

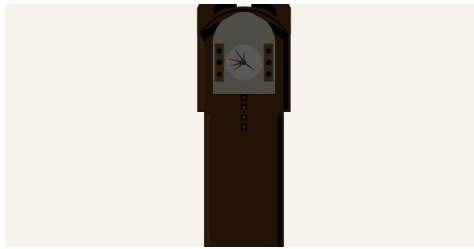
Six round GC9A01 displays (1.28 in, 240 by 240 pixels, SPI), one per person, stand beside the clock face, three on each side. Each has its own ESP32 and joins the same Wi-Fi as the controller, so there are no wires back to the Pi. The content is real recorded footage of each person: a 3 to 5 second loop, cropped to a circle, resized to 240 by 240 and re-encoded as a short frame sequence that fits in the ESP32's flash. A screen also changes state (home, travelling, unknown) on an MQTT command. In the CAD each module is 39.5 mm across with a 32.4 mm active area.



Dial, hands and side screens, from the 3D viewer.

5. Case

A basic outline taken from the reference photos of the grandfather clock Caleb found: a plinth, a hollow trunk, an arched hood with a broken swan-neck pediment and finial, and an arched dial plate. The original movement, weights and pendulum come out. The motor stack runs straight back from the dial, and the dial's brass corner pieces can be reused around a new chapter ring. The outline is a rough shape, about 2 m tall and 495 mm deep (sized to hold the motor stack), not measured from the real case.



Full case from the front, from the 3D viewer.

6. Electronics and parts

Family phones (Find My, with OwnTracks as a fallback) report to Home Assistant, which publishes over MQTT to a Raspberry Pi 4 controller. The Pi does the Haversine math and routing only. It sends one short command per person over USB serial to a Raspberry Pi Pico, which generates all six motors' step pulses in hardware and drives six A4988 stepper drivers from a 12 V supply. The Pi also updates the six ESP32 display nodes over Wi-Fi. Wiring is in the [circuit diagram](#).

Order checklist: <https://austin-hunter1.github.io/HunterClock/order-checklist.html>. It lists every part by order, with prices, sources and order numbers, and has check boxes for tracking what has been bought. Its check marks and notes are saved only in the browser they are made in.

Module	Estimated cost
Controller and networking (Pi 4, Pico, SD card)	\$119 to \$159
Six-shaft hub (6 hollow-shaft motors, drivers, tubing, bearings, dial, hands)	\$305
Six display nodes (ESP32, GC9A01 screen, bezel, cable)	\$82
Power and wiring	\$74
Fabrication and mounting	\$40
Total	\$620 to \$660

Orders placed: Amazon on October 5, 2026 (\$147.00: drivers, ESP32s, screens, Pico, power supplies, capacitors, USB charger, breadboards) and StepperOnline order 330093 on October 6 (all six hollow-shaft motors and the prototype motor, shipped by DHL from China). Still to buy: Pi 4, SD card, extra tubing and bearings, and the fabrication parts.

7. To Do Before Thanksgiving

- Find case dimensions specifically
- Design hands & 3D print (see ambigrams for potential hand internals)
- Record videos of each person for moving picture frame
- Finish code controlling dynamics and updates and visual displays

The case dimensions decide whether the hub fits: hood depth, trunk depth, and the narrowest interior width. The CAD stack is about 420 mm deep and the NEMA23 frame is 57 mm square, the widest part of it.

Files and rebuilding the CAD

Edit `build_cad.py` in the [repository](#) and push to main. The "Build CAD" workflow regenerates the STEP, STL and GLB files, which can be downloaded from the run on the [Actions tab](#). Motor models are in the `motors` folder.